

Week-2 Module - Programming Practice 3rd year

1. Fraud Transaction Detection (Linear Search)

```
Code - #include <iostream>
#include <iostream>
using namespace std;
int main() {
    int N, Key;
    cin >> N;
    int arr[N];
    for (int i=0; i<N; i++) {
        cin >> arr[i];
    }
    cin >> Key;
    bool found = false;
    for (int i=0; i<N; i++) {
        if (arr[i] == Key) {
            found = true;
            break;
        }
    }
    if (found)
        cout << " Fraud Transaction Found";
    else
        cout << " Transaction Not Found";
    return 0;
}
```

PS D:\c++\week2> g++ linearsearch.cpp -o linearsearch

PS D:\c++\week2> .\linearsearch

5 1201 1305 1450 1408 1502 1450

Fraud Transaction Found

PS D:\c++\week2> █

2. ~~Search~~ log Search (Binary Search)

```
#include <iostream>
using namespace std;
int main() {
    int N, target;
    cin >> N;
    int arr[N];
    for (int i = 0; i < N; i++) {
        cin >> arr[i];
    }
    cin >> target;
    int low = 0, high = N - 1;
    bool found = false;
    while (low <= high) {
        int mid = (low + high) / 2;
        if (arr[mid] == target) {
            found = true;
            break;
        }
        else if (arr[mid] < target)
            low = mid + 1;
        else
            high = mid - 1;
    }
    if (found)
        cout << "log found";
    else
        cout << "log Not found";
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\c++\week2> g++ binarysearch.cpp -o binarysearch

PS D:\c++\week2> .\binarysearch

7 1001 1005 1010 1015 1020 1030 1040 1015

Log Found

PS D:\c++\week2> |

3. Sort employee Salaries (Bubble Sort)

```
Code - #include <iostream>
using namespace std;
int main()
{
    int N;
    cin >> N;
    int arr[N];
    for (int i = 0; i < N; i++) {
        cin >> arr[i];
    }
    for (int i = 0; i < N - 1; i++) {
        for (int j = 0; j < N - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                swap(arr[j], arr[j + 1]);
            }
        }
    }
    for (int i = 0; i < N; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell

PS D:\c++\week2> g++ bubblesort.cpp -o bubblesort

PS D:\c++\week2> .\bubblesort

5 45000 32000 55000 28000 60000

28000 32000 45000 55000 60000

PS D:\c++\week2> |

4. Sent website Response time (insertion sent)

```

code - #include <iostream>
using namespace std;
int main() {
    int N;
    cin >> N;
    int arr[N];
    for (int i = 0; i < N; i++) {
        cin >> arr[i];
    }
    for (int i = 1; i < N; i++) {
        int Key = arr[i];
        int j = i - 1;
        while (j >= 0 & arr[j] > Key) {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = Key;
    }
    for (int i = 0; i < N; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}

```

```
PS D:\c++\week2> g++ insertion.cpp -o insertion
```

```
PS D:\c++\week2> .\insertion
```

5 250 120 320 180 100

100 120 180 250 320

```
PS D:\c++\week2>
```

5. Prioritize Bug Reports (selection sort)

```

Code - #include <iostream>
        using namespace std;
        int main() {
            int N;
            cin >> N;
            int arr[N];
            for (int i = 0; i < N; i++) {
                cin >> arr[i];
            }
            for (int i = 0; i < N - 1; i++) {
                int minIndex = i;
                for (int j = i + 1; j < N; j++) {
                    if (arr[j] < arr[
                        minIndex]) {
                        minIndex = j;
                    }
                }
                swap(arr[i], arr[minIndex]);
            }
            for (int i = 0; i < N; i++) {
                cout << arr[i] << " ";
            }
            return 0;
        }

```

```
PS D:\c++\week2> g++ selection.cpp -o selection
```

```
PS D:\c++\week2> .\selection
```

592815

1 2 5 8 9

```
PS D:\c++\week2>
```